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Abstract

Five supersonic and hypersonic, blowdown wind tunnels at NASA Langely are designed for high-Reynolds-number, quiet-flow operation covering a Mach number range from 2.4 to 18. A brief status report on each of the facilities is presented along with new nozzle performance data for the Mach 18 Quiet Helium Tunnel and the Mach 6 Quiet Nozzle in the Nozzle Test Chamber. Additional items discussed in the paper include: experimental transition data on a slender cone with a flared afterbody in the Mach 6 facility under complete quiet flow conditions; nozzle inlet flow-field computations for the Mach 18 helium facility; nozzle coordinate and waviness measurement data for the Mach 8 facility; new Mach 2.4 nozzle design proposals; and an assessment of required quiet tunnel technology research.

Introduction

Development is continuing on a group of low-disturbance or quiet supersonic/hypersonic wind tunnels for instability and transition testing at NASA Langley. This report provides a progress update including discussion of tunnel fabrication, operational characteristics, quiet tunnel technical issues that are outstanding, and data for a Mach 6 transition model. The quiet tunnels under consideration are specifically designed for high Reynolds number operation with very low free-stream-disturbance levels. This capability is required to provide the proper conditions for instability and transition testing at high speeds. The group consists of small-to-medium sized, ideal gas (air, helium), blowdown tunnels covering a Mach number range from 2.4 to 18. The quiet length Reynolds number, $Re_{\Delta X}$, based on free-stream unit Reynolds number and the axial extent of usable quiet flow fall in the overall range of 6×10^6 to 25×10^6. Free-stream disturbance fluctuation levels in the nozzle test regions are in the range of 0.05 percent of the corresponding mean quantities. The Reynolds number range includes both measured values for operational tunnels and theoretical values based on linear stability theory for those under development. Work is underway to extend the quiet length Reynolds number capability range to 80 million in a Mach 2.4 nozzle to support advanced high speed vehicle test requirements, particularly supersonic laminar flow control.

The test capability offered by these facilities is necessary due to possible sensitivity of transition on models to facility-dependent noise at Mach numbers greater than one. The dominant facility noise source at speeds greater than about 2.5 is eddy-Mach-wave radiation emanating from transitional or turbulent boundary layers on the tunnel's nozzle walls (ref. 1). Lesser amounts of noise propagate from the settling chamber but this noise source becomes more important at lower Mach numbers. Quiet tunnel technology aims to eliminate nozzle boundary-layer-induced noise by maintaining laminar flow on the nozzle walls to as high a Reynolds number as possible. Settling chamber noise is controlled with acoustic treatments within the settling chamber itself. References 2 through 17 selectively recount the development of the quiet tunnels from the mid-80's including citations on both tunnel technology and experimental transition data. There is currently no activity outside of NASA Langley in the high Reynolds number, quiet facility area. This is due primarily to the high initial startup costs and the pre-existence of the required infrastructure in the

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NASA Langley Hypersonic Facilities Complex. Several, low Reynolds number, quiet facilities are under development, however, including the Mach 2.5 Low Disturbance Wind Tunnel at NASA-Ames (ref. 18), a Ludwieg tube facility at Purdue University (ref. 19, 20), and the Montana State University Mach 3 Supersonic Wind Tunnel (ref. 21).

Five Langley facilities are specifically designed for high Reynolds number, low-disturbance operation. They are: the Supersonic Low-Disturbance Pilot Tunnel (SLDPT), the Mach 6 Quiet Nozzle in the Nozzle Test Chamber (M6NTC), the Mach 18 Quiet Helium Tunnel (M18QHT), the Mach 8 Variable Density Tunnel (M8VDT), and the High-Speed Low-Disturbance Tunnel (HSLDT). A sixth facility, the 20 Inch Supersonic Wind Tunnel (SWT), the former JPL 20 Inch Supersonic Wind Tunnel, is not designed for high-Reynolds number quiet operation but has had significant new flow quality enhancement features added since its relocation to Langley. These may permit its use in low Reynolds number testing of receptivity and attachment line issues and is therefore listed among the quiet tunnels. A brief status summary of each of the facilities follows along with an appraisal of additional study requirements to advance quiet tunnel technology.

Brief Tunnel Status

The current configuration and operational status of the SLDPT are the same as last reported in reference 9. The facility is fully operational and currently supports tests related to supersonic-laminar-flow-control for the High Speed Civil Transport (HSCT) and basic laminar-flow-instability research. In the FY1995-97 time-frame the tunnel is tentatively scheduled to receive two new, pilot, quiet nozzles. These are designed to demonstrate Mach 2.4 quiet nozzle technology in two different Reynolds number ranges. (The SLDPT and its larger counterpart, the HSLDT, currently have or are designed to use Mach 3.5 nozzles.) The first pilot nozzle will test Mach 2.4 quiet flow capability as a condition for eventual construction of the HSLDT with a Mach 2.4 nozzle. Fabrication of the HSLDT is currently on hold pending the outcome of the Mach 2.4 pilot nozzle study. Its design quiet flow length Reynolds number is 20 million. The second nozzle will attempt to significantly increase the Reynolds number for quiet operation in support of NASA's High Speed Research (HSR) program. Due to the present inability to drill supersonic laminar flow control (SLFC) suction panel holes smaller than those for flight hardware, wind tunnel SLFC testing must be conducted at flight unit Reynolds numbers to achieve proper suction hole Reynolds numbers. A quiet length Reynolds number of 80 million has been selected as an appropriate target for such operation. The second nozzle will explore techniques to reach this high Reynolds number in the scaled conditions available with the SLDPT. The techniques to be used include axisymmetric design, ultra slow expansion contour, high polish, throat region heating, and possibly nozzle cooling. This research will help to lay the foundation for a future very large quiet facility operating at flight unit Reynolds numbers. The two new Mach 2.4 pilot nozzles are presented in detail in a companion paper, reference 22.

Recent nozzle and model transition data acquired in the M6NTC clearly show that the facility is suitable for transition experiments up to a Reynolds number of approximately $6x10^6$ based on the axial extent of usable quiet flow and free-stream unit Reynolds number. While this is only approximately one half the design performance of the facility, it nonetheless demonstrates that transition experiments involving modifiers such as pressure gradient, cooling or cross-flow, which reduce the transition Reynolds number, can be effectively studied in this facility. New hot-wire data showing nozzle laminar flow performance and transition data from a cone with a flared afterbody have been obtained.

The first runs for the M18QHT in its quiet mode have been conducted. Hot-wire power spectral data indicate that the nozzle wall boundary layer is presently laminar only for low stagnation pressures (<200 psia) and that the nozzle's bleed feature is not currently effective. It is conjectured that the problem lies in the geometry of the bleed slot region. Flow field computations in that region have been conducted to help identify problems. Further computations to redesign the inlet geometry are in progress. Limited testing has been conducted to determine the sensitivity of the nozzle boundary layer to the nozzle slot bleed rate. Further tests with variations in the configuration and rating of noise reduction elements (porous plates and screens) in the settling chamber are planned.

Modification of the M8VDT is nearing completion. Fabrication of all major tunnel subassemblies is either complete or in progress including the nozzle, test section and flow filter. The contract for installation of the tunnel controls and the nozzle bleed system should be awarded in the spring of 1994 with initial operational capability (IOC) projected for the summer of 1994.

The SWT is described in reference 23. Briefly, it is a variable Mach number (1.4 to 5.0) blowdown facility with an 18 inch by 24 inch test section. The tunnel is in the final stages of an extensive upgrade and modification program initiated after reference 23. The major improvements include two new flexible nozzle plates, a new test section, a computer-based nozzle-plate position monitoring system and a separate computer-based system for the monitoring and evaluation of the plate-contouring control system. The two-dimensional flexible nozzle and new settling chamber will provide a uniform test core that may allow quiet operation at low Reynolds numbers. The facility can operate over a wide range of stagnation pressures and temperatures, with a maximum Reynolds number of 20 million/foot at a Mach number of 2.88. Shakedown testing is scheduled to begin in the spring of 1994.

Outstanding Technical Issues

For instability and transition work where facility noise sensitivity is important, the Langely high-speed quiet tunnels represent the only test capability extant short of flight tests. Quiet tunnel technology, however, is not yet mature and work remains to be done to optimize performance. Additional research in four areas in particular would enhance understanding of quiet tunnel performance and allow construction of more efficient facilities. They are: nozzle roughness, nozzle wall heat transfer, residual free-stream noise and nozzle corner-flow/side-wall transition.

The role of roughness in the thin shear layer of the throat region is believed to be reasonably well understood in terms of roughness versus transition correlations although the data are somewhat sparse (refs. 3, 5, 24 and 25). Data generally show that the better the polish, the better the quiet flow performance. The criteria typically quoted is $Re_k \leq 10$ or 12 based on experience with the SLDPT Mach 3.5 nozzle. This was obtained from roughness profilimeter readings with corresponding surface micro-photographs. Transition, however, can be caused by isolated roughnesses that statistical samples from a roughness profilimeter can miss. Assessment of scratch or defect depth from planar photographs (which provide large area views of the surface) is also not without problems. The main obstacle to studying roughness requirements in more detail is the difficulty of obtaining reliable measurements of very small surface imperfections ($\sim 10 \times 10^{-6}$ inch) on an area wide basis. Particularly difficulty is encountered in axisymmetric nozzles where the lack of access and three-dimensional surface curvature cause difficulties. New, innovative surface finish diagnostics are required to obtain accurate area-wide surface finish data. With such capability, systematic tests can be performed to better assess the effects of roughness amplitude and microgeometry.

It has long been thought that heating the throat region can enhance the performance of a quiet tunnel by decreasing the local roughness Reynolds number. Recent tests at Montana State University (ref. 26) as well as anecdotal reports from tests in the Langley M8VDT operating as a conventional tunnel in the 1970's point to the efficacy of throat heating. The concept, however, has not been systematically studied in a high-Reynolds number quiet facility. There is also a question as to whether heating acts primarily through the effect on the roughness Reynolds number or whether the boundary-layer mean flow alteration due to the heating is the dominant cause of the observed effects. In reference 24, the opposite case of throat cooling tends to suggest the latter. The new Mach 2.4 pilot nozzles planned for the SLDPT (discussed below) will incorporate throat heating provisions into their design for future study.

Another area that would benefit from additional study is the residual free-stream noise convecting or propagating downstream from the settling chamber into the test region of the nozzle. With settling chamber quieting features such as high density porous plates and screens, this noise is of very low amplitude but high frequency. This noise can play a role in receptivity problems and needs to be accurately measured and characterized in order to use it as an instability input. Hot-wire anemometry has not yet been shown to be effective in characterizing this noise due to electronic noise limitations. New low noise, high-frequency instrumentation is required. Such equipment is now beginning to become available (e.g., constant voltage anemometer, ref. 27) but is in its infancy, requiring extensive evaluation and development. Another approach would be to vary the ratings and configuration of settling chamber quieting elements while performing transition measurements on models in the nozzle under nominally quiet conditions. This type of systematic evaluation has not yet been performed in a quiet tunnel. A related concern stems from the amplitude and frequency content of free-stream disturbances, both in quiet and conventional tunnels, and their effect on transition. Boundary layers instability is influenced by identifiable frequency bands of input disturbances. If the free-stream disturbances are low enough in amplitude such that significant growth amplitudes are not achieved on the body or if the frequencies of the input disturbances are outside the unstable band, is a quiet facility required? This situation arises in high Mach number flows and is discussed in reference 28.

The final area in need of study is the stability of three-dimensional flow in the corners and on the side walls of rectangular or square nozzles. Langley two-dimension nozzles are deliberately made wide in order to move the disturbances associated with the corners and side walls at least as far downstream as the noise from transition on the contoured walls for design conditions. This approach can lead to excessive tunnel size and correspondingly large fabrication and operating costs. An alternative approach is to use square nozzles. This design is more efficient in terms of mass flow requirements while offering the same desirable fabrication/maintenance features as rectangular nozzles. The unknown element is the influence of corner cross-flow on nozzle transition. Analysis of the corner flow problem for several quiet flow nozzles is found in references 21 and 29.

Detailed Discussion

Two of the developmental quiet tunnels, the M6NTC and the M18QHT, are now operational and new tunnel performance data and model transition data (M6NTC only) is presented. In addition, status of the construction of the M8VDT and a summary of plans for new Mach 2.4 nozzles for the SLDPT are presented.

Mach 6 Quiet Nozzle in the Nozzle Test Chamber (M6NTC)

Two significant activities have taken place in the M6NTC. The first was to better document the quiet flow performance of the tunnel. The second was to measure transition on a cone with a flared afterbody (adverse pressure gradient) under both quiet and noisy tunnel conditions as a test of the tunnel's transition testing capability. A summary of the results for each item is presented below.

M6NTC Performance

A single hot wire was located at the exit plane of the nozzle on the nozzle axis to characterize the nozzle noise. Figure 1 shows a sequence of uncalibrated hot-wire signals (constant voltage anemometer) over a range of tunnel stagnation pressures from 80 to 200 psia. The plots show increasing signal intermittency with pressure with the first spikes appearing sporadically at a stagnation pressure of 135 psia. These are acoustic disturbances that follow Mach lines from locally unstable regions at their origins upstream on the nozzle wall. To better characterize the noise, the wire was moved along a radius at the exit plane of the nozzle for several pressures. The same type of qualitative behavior as seen in Figure 1 was observed out to a radius of 3 inches out of a total exit radius of 3.75 inch for all pressures tested. Beyond 3 inches, various disturbances were observed depending on the probe's location with respect to the nozzle wall boundary layer. These observations place transition on the nozzle wall about 4 inches upstream from the nozzle exit at 130 psia. Further mapping of the nozzle disturbance flow field including assessment of run-to-run repeatability is underway and will be reported on at a later date. The maximum useful unit Reynolds number is just below the onset of intermittency in the test region and the goal is to design experiments that can yield useful information while remaining below that threshold. For completely quiet flow at a maximum pressure of 130 psia this corresponds to a quiet flow length Reynolds number of approximately 6×10^6. Allowing a small amount of intermittency in the free-stream, this can be stretched by about 8%, i.e. to 140 psia (see Figure 1). As will be shown in the next section, these values are adequate to obtain transition on an adverse pressure gradient cone under completely quiet conditions.

Hot-Wire Diagnostics

A difficult aspect of the M6NTC (and other tunnels) performance assessment has been electronic noise present in the hot-wire equipment used to measure the free-stream noise. The noise has been both broadband due to amplification and narrowband due to pickup of various laboratory electromagnetic sources such as motors, heater controls, cathode ray tubes, etc.. Such noise tends to obscure significant features of the transition process and makes nozzle flow assessment difficult. For instance, high broadband noise levels can mask intermittent boundary layer transition events such as those shown in Figure 1 and pick-up of narrowband spike-like signals can be confused for boundary layer intermittency. Initial assessment of the nozzle was reported on in reference 12 using a constant current anemometer system with a frequency compensating amplifier.

This system however was only marginally adequate in terms of noise and signal resolution. Additional efforts were made with a constant temperature anemometer (CTA) (ref. 9) but electronic instability and noise pickup were significant problems. The current investigations have used the new constant voltage anemometer (CCA) (ref . 26) with great success. While not commenting on the relative merits of the three competing anemometry systems in terms of sensitivity or frequency response when applied optimally, the CVA system has proved to have the lowest noise and present the least operational problems for the present experiments.

Transition Experiment on a Cone with Adverse Pressure Gradient

An existing cone model with a flared afterbody to induce an adverse pressure gradient was studied to verify the capability of the M6NTC for transition testing. An outline of the cone is shown positioned in the nozzle in Figure 2 with cone details in Figure 3. The cone was hollow and constructed of thin-skinned stainless steel with thermocouples spot welded along a ray on the inner surface. There was a small installation error generating a 0.2 degree pitch angle, but, no detectable yaw angle. The measured and computed equilibrium wall temperature distributions for several repeat runs are shown in Figure 4. The model was rotated in the strut to alternately place the line of thermocouples on the windward or leeward side generated by the 0.2 degree pitch angle. The computations assume laminar flow and zero angle of attack. The general correspondence is seen to be good allowing for the noticeable effect of the 0.2 degree angle of attack and a small amount of scatter. The large deviation between the computed and measured profile starting at just past 15 inches is due to transition. The peak in the profile near the end of the cone is not necessarily a transitional overshoot associated with the transition process. It may be merely the effect of the relatively cold base support structure of the cone pulling down the several thermocouples in the vicinity of the base. Due to the thermal conductivity of the stainless steel skin, the actual start of transition is smeared out and is most likely in the 16-17 inch region. Linear stability computations for this geometry and test conditions indicate an N factor of 7-8 in this region. This shows that the model substantially follows linear stability theory. Temperature data for the cone was not acquired for noisy nozzle conditions (nozzle bleed flow off) at 130 psia, but was obtained for a much lower pressure of 80 psia. As shown in Figure 5, the response of the cone thermocouples is vastly different indicating transition much earlier on the cone. This data clearly shows that the tunnel is capable of providing a sufficiently low noise free stream to yield data consistent with linear stability theory computations.

Future Plans for the M6NTC

Two major follow-on experiments are currently underway in the M6NTC. One is a more complete study of the cone shown in Figure 3 under equilibrium (approximately adiabatic) wall conditions. This study will include expanded coverage of the nozzle mean and noise flow fields, systematic study of transition variation with unit Reynolds number, bluntness, angle-of-attack, hot-wire surveys of the model boundary layer for mean and fluctuating quantities, axial location of the model within the nozzle, and schlieren photography of the visible aft portion of the cone. The second experiment will add cold wall capability (wall temperature down to approximately one-half of free-stream stagnation temperature) to investigate 2nd mode TS instability sensitivity to wall cooling.

<u>Mach 18 Quiet Helium Tunnel (M18QHT)</u>

The M18QHT is currently operational in its quiet-mode configuration and is undergoing performance testing. Current data show laminar flow is attainable only at low stagnation pressures (approximately 200 psia or less). The bleed slot flow, as currently configured, appears to have no discernible effect on the stability of the nozzle wall boundary layer. The settling chamber and bleed slot region is shown in Figure 6a and 6b respectively. Figure 7 shows uncalibrated hot-wire spectra (CVA) taken at the nozzle exit on the centerline at several stagnation pressures with the nozzle slot bleed flow on and off. Noise measured at this position can be traced to a corresponding acoustic origin on the wall located at about 28 inches downstream from the throat. As can be seen, there is a significant reduction in overall energy for the 200 and 202 psia runs compared to runs at pressures that are only about 15 psi higher. Runs at higher pressures (P_0>217) reveal more energy but no change in the general shape of the spectra. Also, spectra obtained at about 212 psia rise slightly higher than the 215 and 217 psia spectra shown and then at about 207 psia the spectrum falls into the electronic background noise. This sequence may be indicative of a transitional overshoot. No significant change can be attributed to the bleed slot flow. The source of the small amount of energy below 10 kHz for the 200 psia runs is not yet known. It may be acoustic disturbances emanating from the settling chamber. Due to the vortex stretching of the nozzle expansion process, vorticity disturbances generated in the settling chamber and traveling with the flow are generally considered to be too attenuated to show up in the free stream disturbance spectrum.

The most troubling finding of the current tests is that the nozzle slot bleed flow does not affect the state of the nozzle wall boundary layer as has been the case for all other Langley quiet tunnel operations. In the case of previous tunnels, the bleed slot flow starts after a large subsonic contraction of the settling chamber flow. In the current case, however, the bleed slot flow starts before the subsonic contraction (Figure 6b) and there is an abrupt change in direction of the nozzle flow at the location of slot inlet lip. It is possible that disturbances generated in this region are obviating any beneficial effect of the bleed flow in removing settling chamber wall disturbances.

A preliminary analysis of the tunnel bleed slot flow was conducted using a multiblock full Navier-Stokes code currently under development at NASA Langley named LARCK. This code solves the Favre averaged form of the Navier-Stokes equations using a Runge-Kutta time advancement scheme and full approximate storage multigrid. The governing equations are discretized using a cell centered finite volume method. The convection terms are treated using a third order upwind biased scheme and the diffusion terms are treated by applying Green's theorem around auxiliary control volumes. Turbulence closure is achieved through the use of the two equation shear stress transport model (ref. 31). The tunnel bleed slot flow, as shown in Figure 8, was computed as being fully turbulent from the inflow boundary on the tunnel outer wall through the bleed slot throat, laminar from the bleed slot leading edge on the inner wall, and laminar from the bleed slot leading edge to the tunnel throat. The computations are for a bleed slot gap of 0.012 inches corresponding to the hot wire spectral data shown in Figure 7.

As can be seen, in Figure 7, the stagnation point of the flow is on the nozzle inlet ramp a short distance downstream from the bleed lip. This indicates that the bleed slot mass flow rate is too high. The results shown are for steady flow. It is extremely unlikely that the stagnation point is stationary in the actual flow. Any unsteadiness would place undesirable disturbances into the nozzle inlet. Also evident in Figure 8 is the large angle through which the inlet flow is turned downstream of the stagnation point. This raises the possibility of Görtler instabilities and unsteady separation bubbles. The computation shows a separation bubble on the bleed side of the nozzle lip, which is likely an unsteady structure that could generate disturbances that feed back to the primary nozzle flow. A smaller bleed gap of 0.008 was previously tested before the computations became available and was also unable to affect the nozzle wall boundary layer state. For this case, the stagnation point would have been closer to the bleed lip. Further study of the bleed slot configuration is clearly necessary. Possible solutions under investigation are reshaping of the bleed lip and reconfiguring the entire bleed slot region geometry to more closely resemble that of other Langley quiet tunnels.

Mach 8 Variable Density Tunnel (M8VDT)

Fabrication of the M8VDT modification hardware (nozzle, model enclosure, bleed system, settling chamber noise control elements) is nearing completion. Actual installation is scheduled to take place during the summer of 1994. The most difficult and important part of the new M8VDT is clearly the nozzle. Due to the length of the nozzle and the stringent coordinate, waviness and finish requirements, the nozzle must be fabricated in five sections to allow use of precision lathes. The five sections are the throat, three extension sections and the model enclosure as shown in Figure 9. The nozzle continues up to the center of the window at X=139.82 inch. The wall of the test enclosure is mildly contoured and part of the nozzle. The throat section itself contains an assembly joint at X= 7.2 inch due to length restrictions of the particular lathe used for the nozzle throat. The overall nozzle fabrication is divided among three fabricators depending upon the size and precision requirements. To insure that the sections mate within specifications, two pairs of precision matched joint gages were fabricated for the joint between the throat and section 1 and the joint between section 3 and the model enclosure.

Two nozzle throats and section 1 extensions are being fabricated. One throat will be machined to within only 0.010-0.020 inch of the final contour and will be used for tunnel start up. It is not possible to guarantee an adequately clean tunnel system on initial startup to insure that no particle damage will occur to the nozzle throat. After the start up runs, the first throat/section-1 combination will become backup equipment. The reason for the two number 1 sections is that it is not possible to build one section 1 that fits both throat sections without incurring a minor step at the joint. Once a throat and section-1 are mated together and run at tunnel operating temperatures, it will not be possible to disassemble and reassemble the joint without causing a step. By having two throat/section-1 pairs permanently mated together, those problems are avoided. Work is beginning on the two number 1 sections. The first of the two throat sections (unpolished version) is complete. Nozzle sections 2,3 and the model enclosure are also complete.

One of the most important issues with the extension sections is surface waviness. Due to the relatively thicker boundary layer in the extension sections than in the throat region, a highly polished (i.e., reflective) finish is not required. Surface waviness, however, can cause noticeable mean flow non-uniformities in an axisymmteric nozzle and must be avoided. Waviness results from inaccuracies in the lathe, tool wear, material non-uniformities, machine vibrations, etc. An objective criterion for waviness was required for contractual specifications. For the current fabrications, it is defined as the deviation of the local surface slope from the design slope at a given X location averaged over an appropriate length.

$$\overline{W} = \overline{\left.\frac{dR}{dX}\right|_{meas}} - \left.\frac{dR}{dX}\right|_{design} \qquad (1)$$

where W is the waviness, R is the local nozzle radius, X is distance along the nozzle streamwise axis and the overscore denotes a streamwise average. Averaging is required to filter out measurement noise and was implemented by fitting a least squares straight line to successive fixed length intervals along the nozzle and assigning the slope of the line to the middle of the interval. Coordinates were acquired at 0.05 inch intervals and groups of 9 contiguous points were used in the averages. The data was incremented by one coordinate point along the nozzle for each new least squares line. Working interactively with the manufacturer of sections 1, 2 and 3, it was determined that a waviness specification of +/- 0.001 based on a 0.45 inch interval was the best that could be achieved and contractually guaranteed with the machining equipment used. This result is based on a measurement methodology by which the difference between the machined nozzle contour and an independently fabricated/verified two dimensional nozzle contour template was determined with electronic sensors accurate to within 0.0001 inch. This procedure allows systematic error in the lathe to be accounted for by adding a correction to the design nozzle coordinates. The contour of the template was verified with a coordinate measuring machine (CMM) with an accuracy of 0.00005 inch. Also, it was found that measurements needed to be made during quiet off-hours in the shop to avoid measurement errors due to shop vibration.

Figures 10a and 10b show the coordinate and waviness results for sections 2 and 3. As mentioned, the averaging length for the waviness is 0.45 inch. A shorter averaging interval will increase the computed waviness; a longer will reduce it. This ambiguity does not present a problem since the hand lapping techniques used to even out the surface tend to remove waviness over a range of wavelengths both shorter and longer than the 0.45 inch interval chosen for the specification. The real question is how the waviness affects the mean flow and ultimately model boundary-layer stability. While computation of the mean flow based on the actual measured coordinate data and subsequent model boundary-layer stability calculations based on the non-uniform flow are within current CFD capability, such computations are not available at this time for the Mach 8 nozzle. Inviscid small perturbation theory applied to the flow past a wavy wall provides some direction. Reference 30 shows that the surface pressure perturbation coefficient, C_{ps}, is directly proportional to the waviness aspect ratio, h/l, or equivalently the surface angle deviation for small perturbations. For this reason, surface angle error minimization is emphasized in this fabrication. As seen in Figure 10a, there is a 0.0025 inch mismatch between section 2 and 3 at X=70 inch joint. This step will be hand lapped to within the 0.001 waviness specification on assembly.

The data shown in Figures 10a and 10b raises the question as to whether a better job could be done in boring out the interior of a large axisymmetric nozzle with current metal machining technology. The current feeling is that while it may be possible the improvement may not be large. The lathe used in the current fabrication is in a class of high-accuracy, large-diameter-capability equipment. Use of a similar lathe built to closer (but practical) tolerances and operated in an environmentally controlled room might yield a factor of two reduction in nozzle waviness. Beyond that, it is necessary to move to air-bearing type lathes. Such equipment, however, is not without problems. Typical air bearing lathes (e.g., diamond turning equipment) have limited axial reach capability and the force required to cut stainless steel tends to displace the air-bearing in its supporting film. The overall judgment appears to be that current nozzle fabrication is near the best available for the type of nozzle and nozzle material (vacuum induction remelt grade Inconel 600) used.

Some of the fabrication problems may be alleviated by the use of a square cross section nozzle rather than the present axisymmetric nozzle (see ref. 29).

Mach 2.4 Pilot Nozzles for the Supersonic Low Disturbance Pilot Tunnel (SLDPT)

The SLDPT is currently outfitted with the second or "new" Mach 3.5 rapid expansion, slotted nozzle discussed in reference 5. The new High-Speed Civil Transport (HSCT) initiative, however, dictates a lower Mach 2.4 capability. Initial studies are underway to install such capability in the SLDPT by way of two new Mach 2.4 pilot nozzles. The first of these new nozzles is similar to the existing Mach 3.5 2D nozzle with contour modifications required to produce Mach 2.4. It also includes a more gradual subsonic approach, a revised bleed lip geometry, and throat heating provisions. This nozzle is primarily a pilot for the High-Speed Low Disturbance Tunnel (HSLDT). The second nozzle is an attempt to greatly increase (by nearly an order of magnitude) nozzle quiet flow performance to enable development of a future, very large, engineering class facility (on the order of a 20 foot diameter test section T&E facility). The design employs an extremely slow expansion contour to limit the Görtler mode, axisymmetric cross-section to avoid corner/side wall induced problems and throat heating to increase roughness tolerance and provisions to experiment with the effect of wall heating/cooling gradients on flow stability. The design of these nozzles is presented in a companion paper, reference 22. Results of the mean flow and stability design computations are reprinted here in Figures 11 and 12. For the two dimensional design, the highest attainable operating pressure based on mass flow rate and upstream piping pressure losses is anticipated to be about 120 psia. This corresponds to a quiet test core length Reynolds number of 11.4 million and is based on Görtler induced transition at a linear stability theory N factor of 9. This performance could be reduced by corner/side-wall disturbances as shown in the figure. Additional design effort has been focused on the nozzle subsonic contraction and bleed lip region to improve the side wall flow. The axisymmetric nozzle requires less mass flow and is expected to reach a stagnation pressure of 200 psia corresponding to a quiet test core length Reynolds number of 80.1 million based on a Görtler N-factor of 9. This theoretical level of performance is possible due to the ultra slow expansion design that limits Görtler mode growth over a long distance. Complete details and further discussion of the pilot nozzles in presented in reference 22.

Concluding Remarks

The foregoing discussion has: provided a brief status of all of the Langley quiet tunnels; presented topics for research to improve quiet flow performance; provided new data from the M6NTC and the M18QHT; presented nozzle fabrication data for the M8VDT; and discussed new Mach 2.4 nozzle ventures for SLDPT. Based on these results, progress is evident. The new, clearer definition of the M6NTC nozzle quiet flow performance has allowed scheduling of two model tests examining the effects of wall curvature (pressure gradient) and wall cooling. These test will for the first time at Langley include hot-wire boundary-layer surveys of mean and fluctuating flow elements. They also directly support companion CFD efforts to compute both the mean flow and boundary-layer stability. The M18QHT results show a range of quiet flow test capability at low Reynolds number and indicate a path toward higher Reynolds number operation by refinements in the nozzle bleed geometry. Manufacturing data from the M8VDT nozzle is very encouraging, showing that the nozzle is being built at or near the limit of conventional manufacturing capability. New Mach 2.4 quiet nozzles are under development for the SLDPT to support the High-Speed Civil Transport and a new class of large quiet tunnels.

The primary use of the Langley quiet tunnels is laminar flow instability and transition research. In the near future there will be a high Reynolds number test capability over a Mach number range from 2.4 to 18 assuming continuing progress in the development of the M8VDT, M18QHT, and the SLDPT. This unique capability will permit acquisition of data under quiet free-stream conditions pertaining to the many of the types of boundary-layer stability modifiers present in new high-speed vehicle designs. These modifiers include pressure gradient, curvature, bluntness, shock waves, angle of attack, roughness/waviness, suction/blowing, and heat transfer. They also provide the necessary test environment for high-frequency, low-noise instrumentation development and cold flow testing of flight data acquisition hardware designs.

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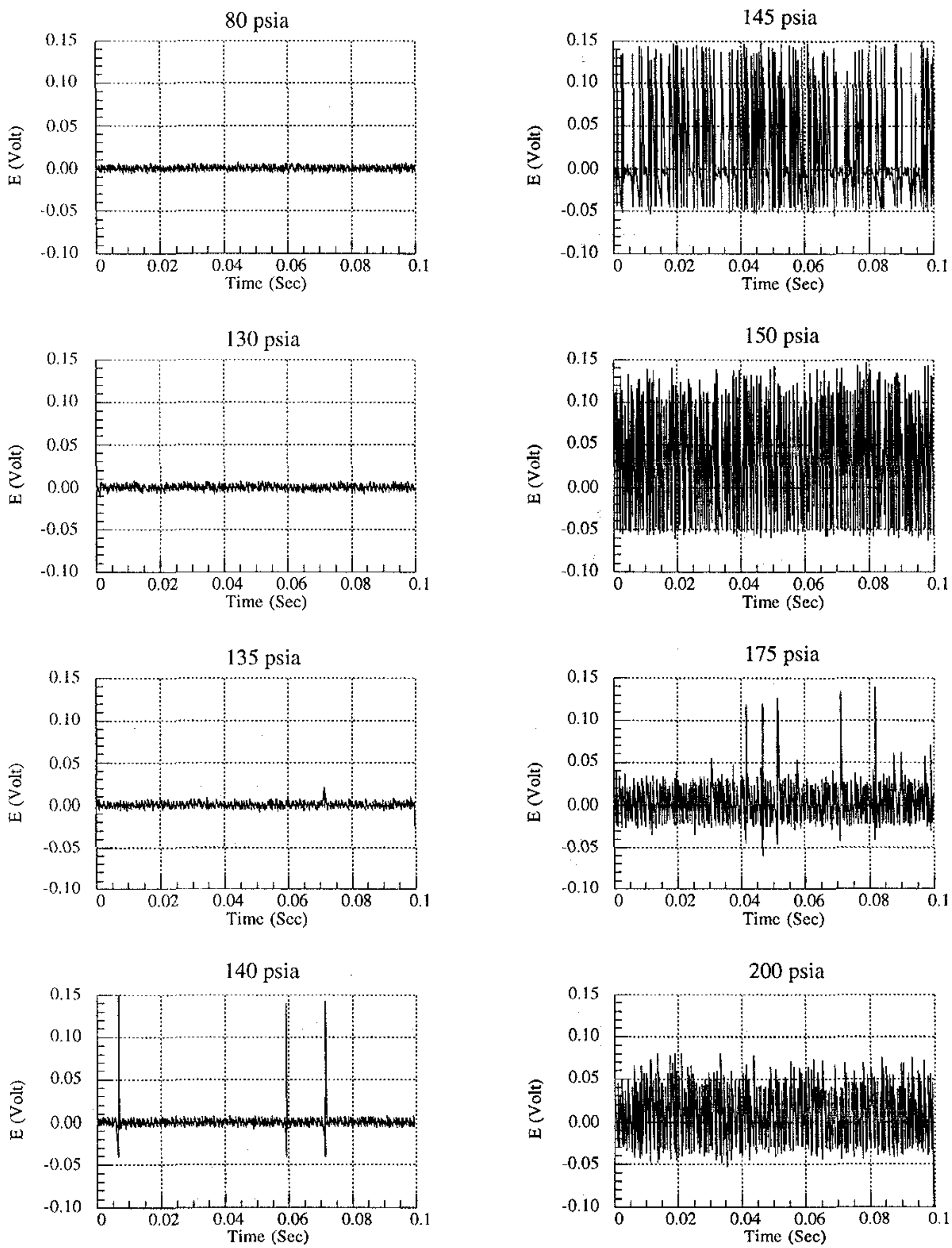

Figure 1. Hot-Wire Signal Traces(Uncalibrated) from Mach 6 Quiet Nozzle Showing Free Stream Noise Development with Increasing Stagnation Pressure. Probe at Nozzle Exit on Centerline.

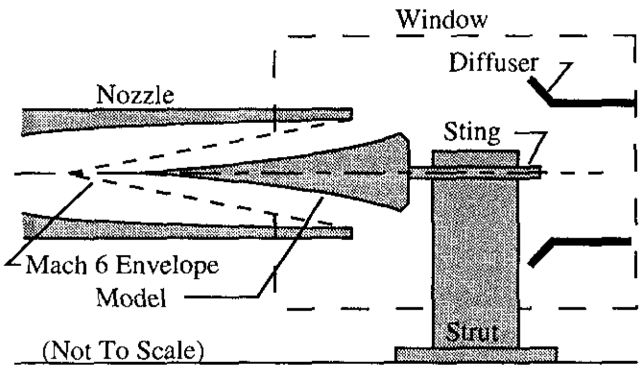

Figure 2. Cone with Flared Afterbody in Mach 6 Quiet Nozzle (M6NTC).

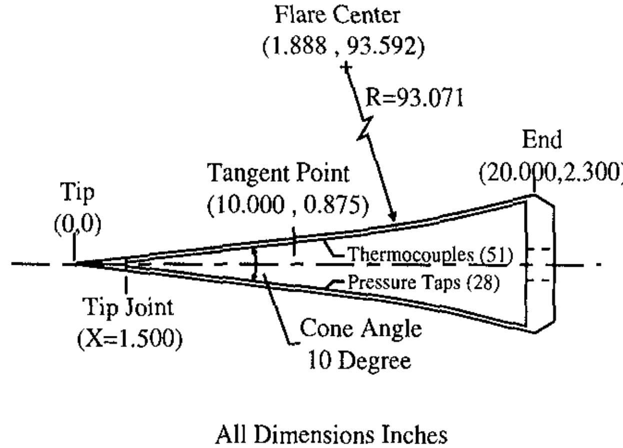

Figure 3. Details of Cone with Flared Afterbody.

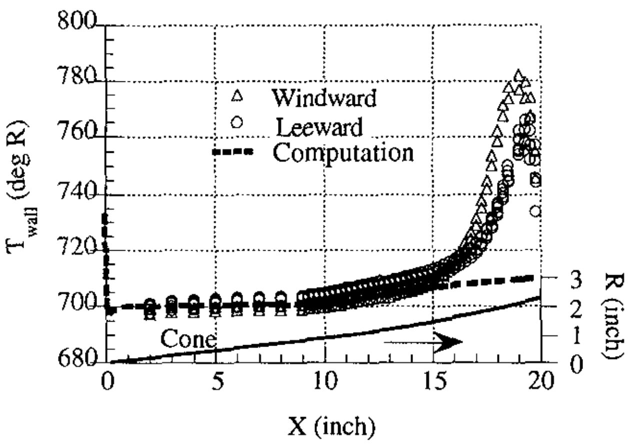

Figure 4. Wall Temperature Distribution for Cone with Flared Afterbody. Stagnation Pressure 130 psia.

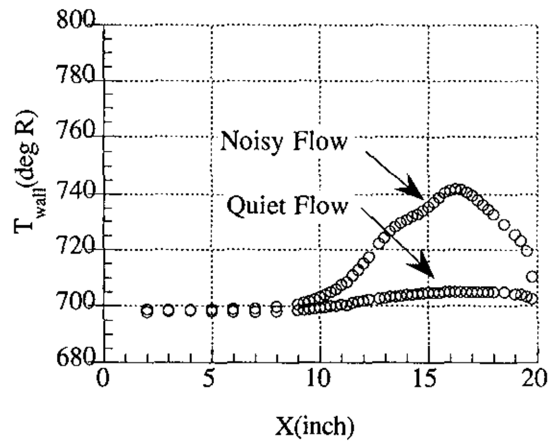

Figure 5. Wall temperature Distribution for Cone with Flared Afterbody. Bleed Slot Flow On (Quiet) and Off (Noisy). Stagnation Pressure 80 psia.

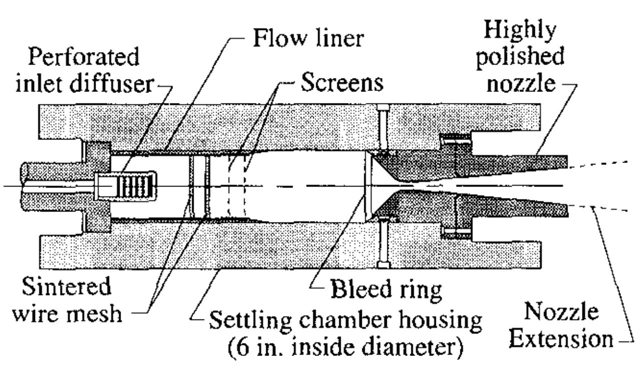

Figure 6a. Mach 18 Quiet Helium Tunnel Settling Chamber.

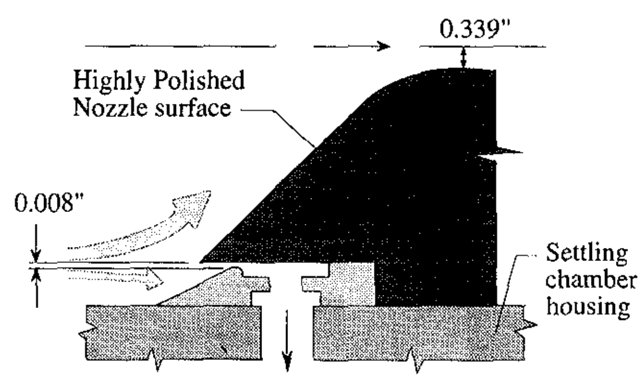

Figure 6b. Enlarged View of Bleed Slot from Figure 6a.

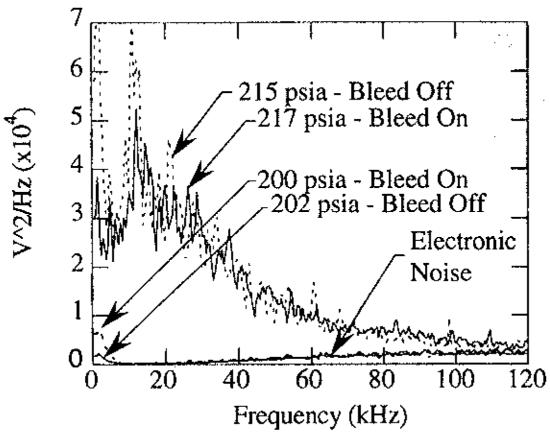

Figure7. Hot-Wire Power Spectra from Mach 18 Quiet Helium Tunnel at Nozzle Exit on Centerline.

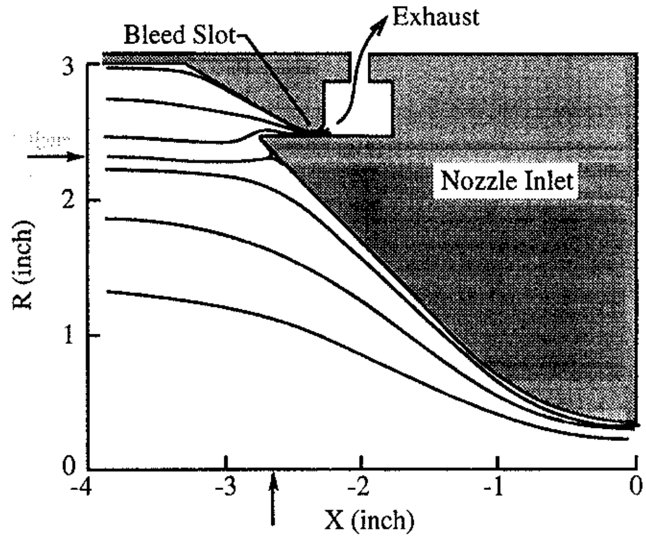

Figure 8. Mach 18 Quiet Helium Tunnel Nozzle Inlet Streamlines. Arrows Indicate Stagnation region. Bleed Gap 0.012 Inch. Stagnation Pressure/Temperature: 485 psia/540 deg R.

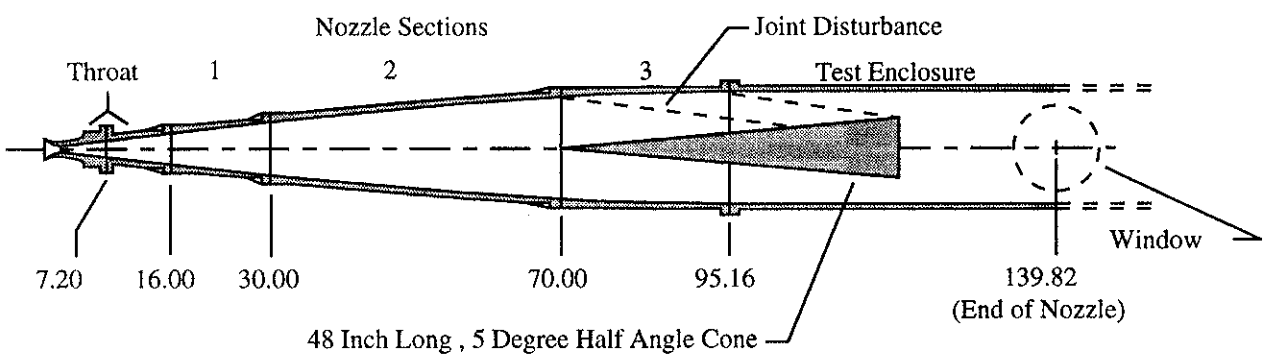

Figure 9. Mach 8 Variable Density Tunnel Nozzle Assembly.

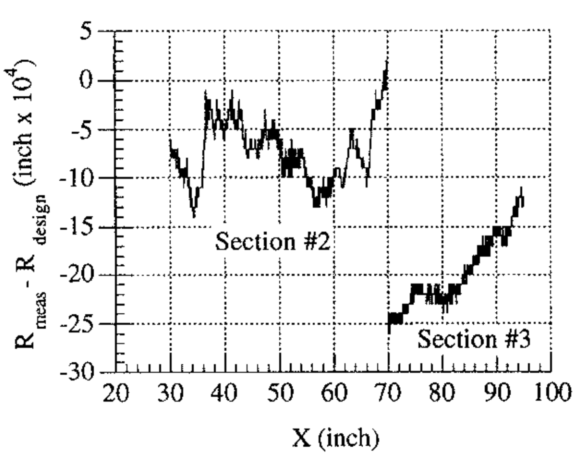

Figure 10a. Mach 8 Variable Density Tunnel Radial Coordinate Deviation From Design Values. Extension Sections 2 and 3.

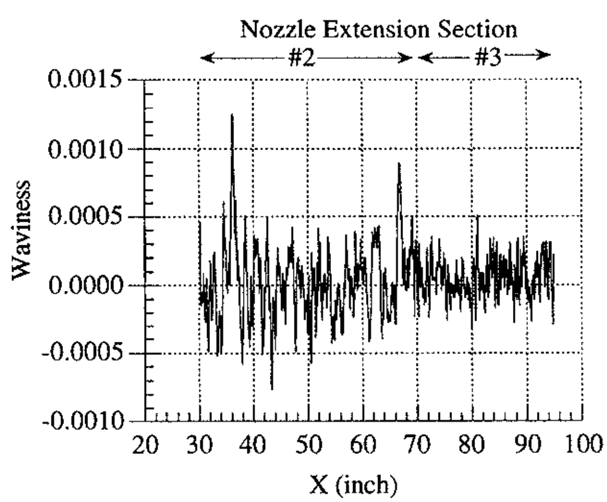

Figure 10b. Nozzle Wall Waviness Computed from Figure 10. Data. Step at X=70 Joint Not Included.

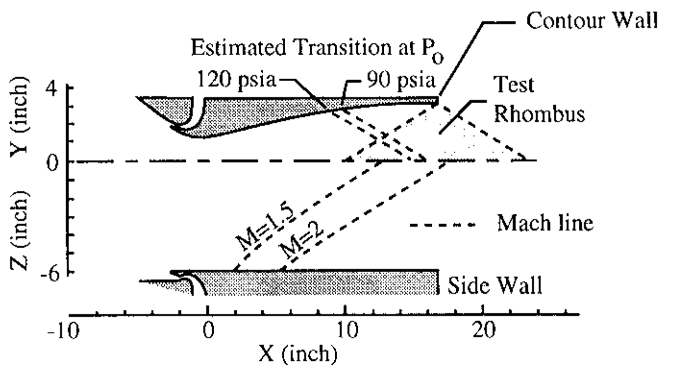

Figure 11. Mach 2.4 Two-Dimensional Pilot Nozzle for Supersonic Low Disturbance Tunnel.

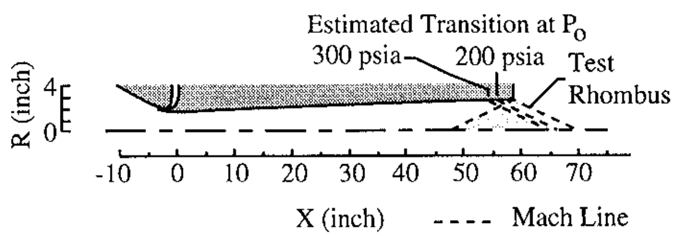

Figure 12. Mach 2.4 Axisymmetric Pilot Nozzle for Supersonic Low Disturbance Tunnel.